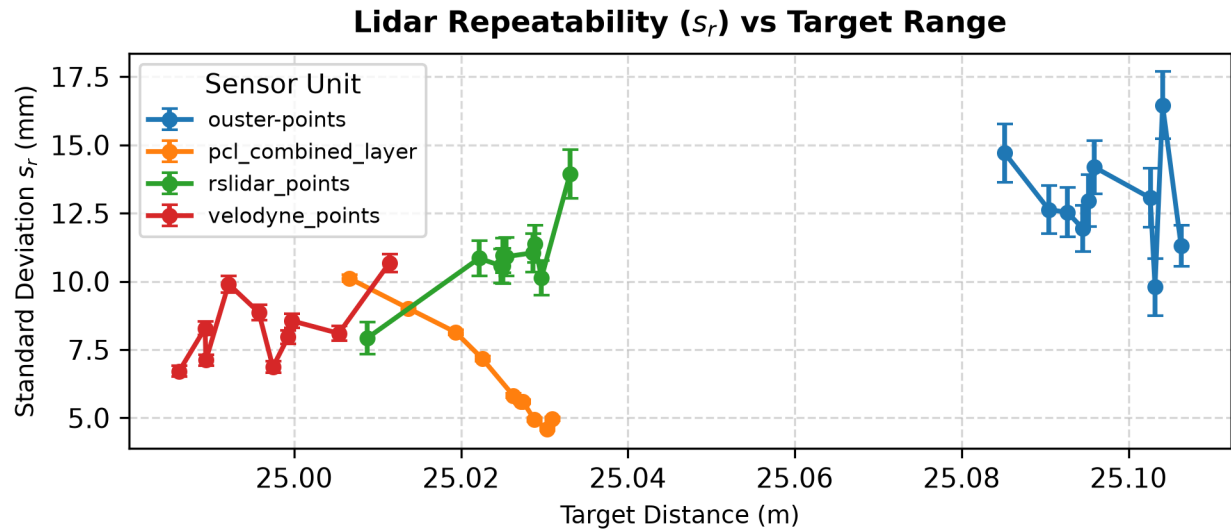


LiDAR Measurement Repeatability Report

Source Summary File: *june_26_2026_laser_rangefinderlidar_repeatability_summary.csv*

Repeatability Precision vs Range



Pooled Sensor Summary

Sensor ID	Trials	Total Points	Avg Distance (m)	Pooled s_r (mm)	Max U_95 (mm)
ouster-points	10	1,812	25.097	13.185	2.460
pcl_combined_layer	10	62,932	25.023	6.967	0.242
rslidar_points	10	2,586	25.025	10.974	1.767
velodyne_points	10	10,565	24.997	8.350	0.674

Target Uncertainty Breakdown

Target	LiDAR	Trial	N	Mean Range (m)	s_r (mm)	u (mm)	U_95 (mm)
target1	ouster-points	1	187	25.0852	14.703	1.075	2.150
target1	ouster-points	3	204	25.0905	12.627	0.884	1.768
target1	ouster-points	10	192	25.0927	12.536	0.905	1.809
target1	ouster-points	9	201	25.0945	11.944	0.842	1.685
target1	ouster-points	8	183	25.0952	12.957	0.958	1.916
target1	ouster-points	2	210	25.0959	14.187	0.979	1.958
target1	ouster-points	6	146	25.1026	13.056	1.081	2.161
target1	ouster-points	5	88	25.1032	9.792	1.044	2.088
target1	ouster-points	4	179	25.1041	16.457	1.230	2.460
target1	ouster-points	7	222	25.1063	11.305	0.759	1.518

target1	pcl_combined_layer	1	7008	25.0066	10.117	0.121	0.242
target1	pcl_combined_layer	8	6749	25.0137	9.009	0.110	0.219
target1	pcl_combined_layer	10	7370	25.0194	8.136	0.095	0.190
target1	pcl_combined_layer	9	6243	25.0225	7.178	0.091	0.182
target1	pcl_combined_layer	7	5712	25.0263	5.817	0.077	0.154
target1	pcl_combined_layer	3	5421	25.0272	5.596	0.076	0.152
target1	pcl_combined_layer	2	6223	25.0274	5.608	0.071	0.142
target1	pcl_combined_layer	6	5417	25.0288	4.942	0.067	0.134
target1	pcl_combined_layer	4	7256	25.0303	4.591	0.054	0.108
target1	pcl_combined_layer	5	5533	25.0310	4.961	0.067	0.133
target1	rslidar_points	1	187	25.0088	7.924	0.579	1.159
target1	rslidar_points	10	275	25.0222	10.848	0.654	1.308
target1	rslidar_points	8	287	25.0248	10.573	0.624	1.248
target1	rslidar_points	6	304	25.0250	10.948	0.628	1.256
target1	rslidar_points	4	267	25.0250	10.572	0.647	1.294
target1	rslidar_points	5	244	25.0254	10.909	0.698	1.397
target1	rslidar_points	2	248	25.0286	11.045	0.701	1.403
target1	rslidar_points	7	275	25.0289	11.378	0.686	1.372
target1	rslidar_points	9	250	25.0296	10.131	0.641	1.281
target1	rslidar_points	3	249	25.0331	13.939	0.883	1.767
target1	velodyne_points	4	1116	24.9862	6.712	0.201	0.402
target1	velodyne_points	6	1106	24.9893	8.285	0.249	0.498
target1	velodyne_points	3	1197	24.9894	7.113	0.206	0.411
target1	velodyne_points	8	1038	24.9922	9.904	0.307	0.615
target1	velodyne_points	7	1025	24.9958	8.866	0.277	0.554
target1	velodyne_points	5	1090	24.9975	6.864	0.208	0.416
target1	velodyne_points	10	1051	24.9993	7.958	0.245	0.491
target1	velodyne_points	2	1051	24.9997	8.562	0.264	0.528
target1	velodyne_points	1	888	25.0054	8.097	0.272	0.543
target1	velodyne_points	9	1003	25.0114	10.675	0.337	0.674